

PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 7 Exam Notes: Fundamentals of Video Coding

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Contents

1	Inter-Frame Redundancy	2
2	Motion Estimation and Compensation	2
2.1	Block Matching Algorithm (BMA)	2
2.2	Search Strategies	2
3	Frame Classification and Sequence Hierarchy	3
4	Hybrid Video Encoder and Standards	3
4.1	Video Standards	3
5	Worked Examples	4

1. Inter-Frame Redundancy

A video sequence is a series of consecutive images (frames) captured over time.

Temporal Redundancy

Inter-Frame (Temporal) Redundancy: Refers to the massive statistical correlation between successive frames. Since video frames are captured at high rates (typically 24 to 60 frames/sec), adjacent frames contain nearly identical background and object structures, with only minor spatial displacements.

2. Motion Estimation and Compensation

Motion estimation detects displacements of objects between frames, while motion compensation encodes only the difference (prediction residual) and the motion displacement vectors.

2.1 Block Matching Algorithm (BMA)

The standard method for motion estimation.

- The current frame is divided into non-overlapping macroblocks of size $N \times N$ (typically 16×16).
- For each block, we search for the best matching block in a reference frame (past or future) within a search range $[-p, p]$.
- **Search Metrics:** Used to measure block similarity. The most common is the **Mean Absolute Difference (MAD)**:

$$\text{MAD}(i, j) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} |F_t(x, y) - F_{t-1}(x + i, y + j)| \quad (1)$$

where F_t is the current block and F_{t-1} is the reference block shifted by displacement (i, j) (Motion Vector).

2.2 Search Strategies

- **Full Search (Exhaustive Search):** Evaluates MAD at every possible integer pixel shift within $[-p, p]$. Guarantees finding the global minimum, but is computationally extremely heavy.
- **Three-Step Search (TSS):** A fast, logarithmic search strategy:
 - a. Start at center, evaluate 9 search locations with a coarse step size $S = \lfloor p/2 \rfloor$.
 - b. Move center to the location yielding minimum MAD. Reduce step size to $S = S/2$.
 - c. Evaluate 8 new locations around the new center. Repeat until step size is 1.

Computational savings: Requires only 25 computations for $p = 7$, compared to 225 for Full Search.

3. Frame Classification and Sequence Hierarchy

Video coding standards classify frames into three types to optimize prediction efficiency:

- **I-Frames (Intra-Coded):** Coded independently of other frames using still-image compression (like JPEG). Acts as random access points. Low compression.
- **P-Frames (Predicted):** Coded using forward motion prediction from a previous reference frame (I or P). Moderate compression.
- **B-Frames (Bidirectional Predictive):** Coded using both past and future reference frames (forward and backward prediction). Yields the highest compression ratios.

Video Sequence Structure

Group of Pictures (GOP): The sequence structure of I, P, and B frames. A typical GOP pattern is:

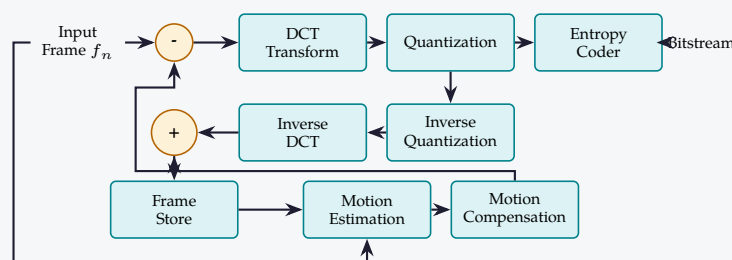
I B B P B B P B B I

Hierarchy: Video Session $\rightarrow$ GOP $\rightarrow$ Picture/Frame $\rightarrow$ Slices $\rightarrow$ Macroblocks (16×16) $\rightarrow$ Blocks (8×8).

4. Hybrid Video Encoder and Standards

Modern video compression standards use a hybrid framework combining DPCM predictive loop (motion estimation) with spatial transform coding (2D DCT).

Hybrid Video Encoder Block Diagram



4.1 Video Standards

- **MPEG-1 / MPEG-2:** Formats used for VCDs (*MPEG* – 1) and DVDs/Digital TV (*MPEG* – 2).
- **H.264/AVC:** Incorporates intra-frame spatial prediction, quarter-pixel motion vector accuracy, and variable block sizes (16×16 down to 4×4).
- **H.265/HEVC:** Replaces macroblocks with Coding Tree Units (CTUs) of size up to 64×64 , reducing bitrates by 50% compared to H.264 for equivalent quality.

5. Worked Examples

Motion Vector Estimation

Problem: Given a 2×2 macroblock in the current frame F_t and its corresponding search region in the reference frame F_{t-1} . Calculate the Mean Absolute Difference (MAD) for candidate displacement vectors $\mathbf{v}_1 = (0, 0)$ and $\mathbf{v}_2 = (0, 1)$, and identify the optimal motion vector.

Current Block F_t :

$$F_t = \begin{bmatrix} 80 & 85 \\ 70 & 75 \end{bmatrix}$$

Reference Search Region F_{t-1} :

	$x = -1$	$x = 0$	$x = 1$
$y = -1$	78	82	88
$y = 0$	72	80	84
$y = 1$	65	71	76

Solution:

(a) For $\mathbf{v}_1 = (0, 0)$: The candidate block in F_{t-1} starting at $(0, 0)$ is:

$$\begin{bmatrix} 80 & 84 \\ 71 & 76 \end{bmatrix}$$

Compute the absolute differences:

$$\text{Diff} = \begin{bmatrix} |80 - 80| & |85 - 84| \\ |70 - 71| & |75 - 76| \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\text{MAD}(0, 0) = \frac{0 + 1 + 1 + 1}{4} = \frac{3}{4} = 0.75$$

(b) For $\mathbf{v}_2 = (0, 1)$ (**vertical shift by $j = 1$**): The candidate block in F_{t-1} shifted by $(0, 1)$ is:

$$\begin{bmatrix} F_{t-1}(0, 1) & F_{t-1}(1, 1) \\ F_{t-1}(0, 2) & F_{t-1}(1, 2) \end{bmatrix} = \begin{bmatrix} 71 & 76 \\ \text{N/A} & \text{N/A} \end{bmatrix}$$

Since this candidate block contains out-of-bounds elements (N/A), we cannot calculate a valid MAD for $\mathbf{v}_2 = (0, 1)$.

Alternative Candidate Analysis: Let us analyze displacement $(i, j) = (-1, -1)$ (shifted top-left by 1 pixel) as an alternative candidate:

- The candidate block in F_{t-1} for $(i, j) = (-1, -1)$ is:

$$\begin{bmatrix} F_{t-1}(-1, -1) & F_{t-1}(0, -1) \\ F_{t-1}(-1, 0) & F_{t-1}(0, 0) \end{bmatrix} = \begin{bmatrix} 78 & 82 \\ 72 & 80 \end{bmatrix}$$

- Compute the absolute differences with the current block F_t :

$$\text{Diff} = \begin{bmatrix} |80 - 78| & |85 - 82| \\ |70 - 72| & |75 - 80| \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix}$$

- Calculate the MAD:

$$\text{MAD}(-1, -1) = \frac{2 + 3 + 2 + 5}{4} = \frac{12}{4} = 3.00$$

Conclusion and Motion Vector Selection: Comparing the calculated MAD values:

- $\text{MAD}(0, 0) = 0.75$
- $\text{MAD}(-1, -1) = 3.00$

Since $\text{MAD}(0, 0) < \text{MAD}(-1, -1)$, the motion vector $\mathbf{v} = (0, 0)$ is selected as the preferred motion vector.

Quick Revision — Module VII

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Key Formulas at a Glance

Mean Absolute Difference	$MAD(i, j) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} F_t(x, y) - F_{t-1}(x + i, y + j) $
Mean Squared Error	$MSE(i, j) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} [F_t(x, y) - F_{t-1}(x + i, y + j)]^2$
Search Complexity (Full)	$N_{\text{eval}} = (2p + 1)^2 \quad (p = \text{search range})$
Search Complexity (TSS)	$N_{\text{eval}} \approx 9 + 8 \log_2(p)$
Motion Compensation Error	$e_t(x, y) = F_t(x, y) - F_{t-1}(x + dx, y + dy)$

One-Line Definitions

- **Motion Estimation:** The process of locating the displacement vector of a macroblock between successive video frames.
- **Motion Compensation:** The process of reconstructing a frame by shifting reference frame pixels using motion vectors.
- **I-Frame:** An intra-coded frame compressed independently using spatial details, acting as a random access sync point.
- **P-Frame:** A predicted frame coded using forward temporal prediction from a single preceding reference frame.
- **B-Frame:** A bidirectional frame coded using both past and future reference frames, offering maximum compression.
- **GOP:** Group of Pictures, defining the recurring sequence pattern of I, P, and B frames in a video stream.

Frequently Asked / Viva Questions

1. **What is temporal redundancy in video coding?**
It is the high correlation between consecutive frames in a video sequence, where background and moving objects remain mostly identical from frame to frame.
2. **How does the Hybrid Video Encoder combine spatial and temporal compression?**
It uses motion estimation (temporal prediction) to generate a residual prediction error, which is then compressed using 2D DCT and quantization (spatial transform coding).
3. **Why do B-frames achieve much higher compression than P-frames?**
Because B-frames use bidirectional prediction, meaning they can reference macroblocks from both past and future frames, yielding smaller prediction errors.
4. **What is the trade-off of using a large number of B-frames in video streaming?**

Frequently Asked / Viva Questions (contd.)

B-frames require future frames to be decoded first, which introduces frame buffering delay (latency) and increases decoder memory requirements.

5. Explain the Three-Step Search (TSS) algorithm.

It is a fast search method that evaluates 9 points using a coarse step size, shifts the center to the minimum error point, halves the step size, and repeats until the step size is 1.

6. What is a macroblock in video standards?

A macroblock is a 16×16 pixel region of luminance (Y) along with the corresponding chrominance (U, V) blocks, representing the basic processing unit for motion compensation.

7. State the main advantage of H.264/AVC over MPEG-2.

H.264/AVC achieves comparable visual quality at approximately half the bitrate of MPEG-2 by using intra-prediction, multi-frame references, and variable block sizes.

8. What are CTUs in H.265/HEVC?

Coding Tree Units (CTUs) replace macroblocks in H.265. They can be dynamically partitioned into Coding Units (CUs) from 64×64 down to 8×8 , optimizing compression for high-resolution (4K/8K) video.

9. Why are P-frames and B-frames unsuitable as random access entry points?

Because they contain temporal prediction residuals, meaning they cannot be decoded without first decoding their past and future reference frames.

10. Define the Motion Vector.

The displacement vector (dx, dy) representing the horizontal and vertical offset of a matching block in the reference frame relative to the current frame.

Comparison of Video Frame Types

Attribute	I-Frame (Intra)	P-Frame (Predicted)	B-Frame (Bidirectional)
Coding Source	Self-contained (spatial)	Temporal (forward)	Temporal (bidirectional)
Compression Ratio	Low (2 : 1 to 5 : 1)	Moderate (10 : 1 to 20 : 1)	High (30 : 1 to 50 : 1)
Random Access Point	Yes	No	No